

THEORY AND METHODS
A COMPARISON OF GENERALIZED RIDGE
REGRESSION ESTIMATORS THROUGH
ESTIMATES OF THEIR MEAN SQUARED
ERRORS

D.O. Chalton

Department of Statistics, University of the Western Cape, Private Bag X17,
Bellville 7535

C.G. Troskie

Department of Statistical Sciences, University of Cape Town, Rondebosch
7700

Key words: Mean squared errors; ridge regression estimators.

Summary: A comparison is made between the ridge estimators of Hoerl and Kennard (1970) and Troskie and Chalton (1996) with respect to the unbiased estimates of their mean squared errors.

1. Introduction

A number of ridge regression estimators have been proposed to deal with the problem of multicollinear data. Various properties of these estimators have been described and compared. Troskie and Chalton (1996) proposed a Bayesian estimate for the constants in ridge regression. In a comparison with other estimators, they showed that their ridge regression estimator has superior efficiency and less bias relative to least squares; the ridge regression estimator of Hoerl and Kennard (1970) being the next best estimator.

Srivastava and Giles (1991) have derived an unbiased estimator of the mean squared error of the ridge regression estimator of Hoerl and Kennard (1970). In this paper we obtain an unbiased estimator of the mean squared error of the ridge regression estimator of Troskie and Chalton (1996), and a comparison is made of the precision of the two ridge estimators.

2. The model

We consider the regression model

$$y = X\beta + \epsilon, \quad (2.1)$$

where y is an n -dimensional column vector of responses, X is an $(n \times p)$ matrix of known constants, β is a p -dimensional column vector of regression coefficients and ϵ follows a p -variate normal distribution with mean vector 0 and variance matrix $\sigma^2 I$.

We standardize the X matrix so that $X'X$ is in correlation form. The ordinary least squares estimate of β is given by

$$\begin{aligned} \hat{\beta} &= (X'X)^{-1}X'y \\ \text{and } \hat{\sigma}^2 &= s/v, \end{aligned}$$

where $s = (y - X\hat{\beta})'(y - X\hat{\beta})$ and $v = n - p$.

When the matrix X is nearly multicollinear, the matrix $X'X$ will be close to singular. The ridge estimator of β is defined by perturbing the matrix $X'X$ away from singularity, i.e. '

$$\hat{\beta}(k_i) = (X'X + K)^{-1}X'y$$

where $K = \text{diag}(k_1, \dots, k_p)$ and the k_i are called ridge constants.

Let P be an orthogonal matrix such that $P'X'XP = \Lambda = \text{diag}(\lambda_i)$. Then model 2.1 can be rewritten as

$$y = X\beta + \epsilon = (XP)(P'\beta) + \epsilon = Z\alpha + \epsilon$$

with $Z'Z = \Lambda$. The ordinary least squares estimate of α is

$$\hat{\alpha} = (Z'Z)^{-1}Z'y,$$

and the ridge estimate is

$$\hat{\alpha}(k_i) = (Z'Z + K)^{-1}Z'y, \text{ with } \hat{\alpha}_i(k_i) = \left(\frac{\lambda_i}{\lambda_i + k_i} \right) \alpha_i.$$

Hoerl and Kennard (1970) proposed the ridge estimator

$$\hat{\alpha}_i^*(k_i) = \left[\frac{\lambda_i \hat{\alpha}_i^2}{\lambda_i \hat{\alpha}_i^2 + s/v} \right] \hat{\alpha}_i, \quad (2.2)$$

where the ridge constants are estimated by

$$\hat{k}_i = \frac{s/v}{\hat{\alpha}_i^2}.$$

The Troskie and Chalton (1996) ridge estimator is

$$\hat{\alpha}_i^*(k_i) = \left[\frac{\lambda_i \hat{\alpha}_i^2 + s/v}{\lambda_i \hat{\alpha}_i^2 + 2s/v} \right] \hat{\alpha}_i, \quad (2.3)$$

and the ridge constants are estimated by

$$\hat{k}_i = \frac{s/v}{\hat{\alpha}_i^2 + s/v\lambda_i}.$$

3. Mean squared error

Srivastava and Giles (1991) have obtained an unbiased estimator of the bias of $\hat{\alpha}_i^*(k_i)$ as

$$\hat{B}(\hat{\alpha}_i^*(k_i)) = - \left[\frac{\hat{\alpha}_i}{t_i^2 + 1} \right],$$

where $t_i = \hat{\alpha}_i(\lambda_i v/s)^{\frac{1}{2}}$ is the t -ratio for testing if $\alpha_i = 0$.

It is easy to show that an unbiased estimate of the bias of $\hat{\alpha}_i^*(k_i)$ is

$$\hat{B}(\hat{\alpha}_i^*(k_i)) = - \left[\frac{\hat{\alpha}_i}{t_i^2 + 2} \right].$$

For the Hoerl and Kennard (1970) ridge estimator, an unbiased estimate of the mean squared error is obtained by Srivastava and Giles (1991) as

$$\hat{M}(\hat{\alpha}_i^*(k_i)) = \left(\frac{s}{v\lambda_i} \right) \left[1 + \frac{t_i^2}{(t_i^2 + 1)^2} + v \int_0^1 \frac{(t_i^2 - x)x^{\frac{v}{2}}}{(t_i^2 + x)^2} dx \right].$$

Using the approach of Srivastava and Giles (1991) it is easy to show that for the Troskie and Chalton (1996) ridge estimator, an unbiased estimator of the mean squared error is

$$\hat{M}(\hat{\alpha}_i^*(k_i)) = \left(\frac{s}{v\lambda_i} \right) \left[1 + \frac{t_i^2}{(t_i^2 + 2)^2} + v \int_0^1 \frac{(t_i^2 - 2x)x^{\frac{v}{2}}}{(t_i^2 + 2x)^2} dx \right].$$

To compare the estimated mean squared errors of the two ridge estimators we have computed values of the ratio $\hat{M}(\hat{\alpha}_i^*(k_i))/\hat{M}(\hat{\alpha}_i^*(k_i))$ and these are presented in Table 1 for some values of v and t_i .

Table 1. Values of $\hat{M}(\hat{\alpha}_i^*(k_i))/\hat{M}(\hat{\alpha}_i^*(k_i))$

t_i	0.5	0.6	0.8	1.0	1.2	1.4	1.6	1.8	2.0	3.0	5.0
v											
1	1.20	1.05	0.98	0.94	0.92	0.91	0.91	0.91	0.92	0.94	0.97
2	1.40	1.14	1.02	0.96	0.93	0.91	0.91	0.91	0.91	0.93	0.97
4	1.64	1.24	1.07	0.98	0.94	0.92	0.91	0.90	0.91	0.93	0.96
6	1.76	1.30	1.10	1.00	0.95	0.92	0.91	0.90	0.91	0.92	0.96
8	1.82	1.34	1.12	1.01	0.95	0.93	0.91	0.90	0.91	0.92	0.96
10	1.86	1.36	1.14	1.02	0.96	0.93	0.91	0.90	0.91	0.92	0.96
20	1.94	1.40	1.16	1.04	0.97	0.94	0.92	0.91	0.91	0.92	0.96
30	1.97	1.42	1.17	1.05	0.98	0.94	0.92	0.91	0.90	0.92	0.96

The table shows that for almost all values of t_i and v the Troskie and Chalton (1996) estimator has lower estimated mean squared error than the

Hoerl and Kennard (1970) estimator. Note that for values of t_i less than 0.5, $\hat{M}(\hat{\alpha}_i^*(k_i))$ can take on negative values.

4. A class of estimators

If we consider the expressions (2.2) and (2.3) for the ridge estimators and substitute t_i for $\hat{\alpha}_i(\lambda v/s)^{\frac{1}{2}}$, it is apparent that the two estimators are members of a class of estimators defined by

$$\hat{\alpha}_i(k_i) = \left[\frac{t_i^2 + c}{t_i^2 + c + 1} \right] \hat{\alpha}_i,$$

where $c \geq 0$. Thus when $c = 0$, we obtain the Hoerl and Kennard (1970) estimator, and when $c = 1$ we obtain the Troskie and Chalton (1996) estimator. Which estimator in this class would be preferred? Using the unbiased estimate of the mean squared error as a criterion, given by

$$\bar{M}(\hat{\alpha}_i(k_i)) = \left(\frac{s}{v\lambda_i} \right) \left[1 + \frac{t_i^2}{(t_i^2 + c + 1)^2} + v \int_0^1 \frac{(t_i^2 - (c+1)x)x^{\frac{c}{2}}}{(t_i^2 + (c+1)x)^2} dx \right],$$

and assuming standardisation such that $(s/v\lambda_i) = 1$, then Figure 1 shows the relation between the estimate of the mean squared error and t_i^2 , when $v = 30$ for some values of c . The solid line represents the Troskie and Chalton (1996) estimator, and we suggest that this is the preferred estimator as the only estimator with zero mean squared error when t_i^2 is zero.

Acknowledgement

The authors wish to thank an anonymous referee for helpful comments and suggestions.

References

- HOERL, A.E. & KENNARD, R.W. (1970). Ridge regression. Biased estimation for non-orthogonal problems. *Technometrics*, **12**, 55–67.

SRIVASTAVA, VK. & GILES, D.E.A. (1991). Unbiased estimation of the mean squared error of the feasible generalised ridge regression estimator. *Commun. Statist.-Theory Meth*, **20**, 2375–2386.

TROSKIE, C.G. & CHALTON, D.O. (1996). A Bayesian estimate for the constants in ridge regression. *South African Statistical Journal*, **30**, 119–137.

